

An Introduction to Particle Filters

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Aim of Session

- Introduce State Space Models.
- Derive the filtering equations for state space models.
- Show how the Bootstrap Filter can be used for inference.
- Understand the Monte Carlo error of the Bootstrap Filter.
- Discuss some ways of improving the basic Bootstrap Filter.
- Give pointers to the wider class of particle filters.

State Space Models

Particle filters allow for inference for **state space models**: models with

- an unobserved **Markov process** $X_1, X_2, \dots, X_t, \dots$; and
- partial observations $Y_1, Y_2, \dots, Y_t, \dots$, such that observation Y_t only depends on X_t (formally it is conditionally independent of the other X_s values given X_t).

We are interested in estimating the states, for example estimating X_t given $Y_{1:t} = \{Y_1, \dots, Y_t\}$.

Generally models will have unknown parameters **but we will assume all parameters are known**.

(Particle filters struggle with estimating parameters – in general you need to run them for different values.)

State Space Models

Mathematically we can define a state space model via:

$$X_1 \sim p(x_1),$$

$$X_t | X_{1:t-1} \sim p(x_t | x_{t-1}),$$

$$Y_t | X_{1:t}, Y_{1:t-1} \sim p(y_t | x_t),$$

where we use p for a general probability density (mass) function; and use the arguments to indicate which.

This leads to a simple way of simulating the state-space model.

In practice these densities may depend on parameters (but we are assuming these are known/conditioned on).

Filtering, Prediction and Smoothing

Particle filters are used to make inference about the latent states given the observations to date, $y_{1:t}$:

- **Filtering:** $p(x_t|y_{1:t})$. (Current state)
- **Prediction:** $p(x_T|y_{1:t})$, for $T > t$. (Future states)
- **Smoothing:** $p(x_T|y_{1:t})$, for $T < t$. (Past states)
- **Inference:** for the parameter θ , e.g. by calculating the likelihood, $p(y_{1:t}|\theta)$.

They are based on calculating these (particularly the filtering density) **recursively**.

Filtering Recursions

We can recursively calculate $p(x_t|y_{1:t})$ from $p(x_{t-1}|y_{1:t-1})$ and y_t by solving the filtering recursions:

$$p(x_t|y_{1:t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|y_{1:t-1})dx_{t-1} \quad (\text{Prediction}).$$

$$p(x_t|y_{1:t}) \propto p(x_t|y_{1:t-1})p(y_t|x_t) \quad (\text{Bayes' rule}),$$

$$p(y_t|y_{1:t-1}) = \int p(x_t|y_{1:t-1})p(y_t|x_t)dx_t \quad (\text{Normalising Constant})$$

These rely on the Markov property of the latent process and the conditional independence property of the observations.

Can be solved analytically for linear-Gaussian models ([Kalman Filter](#)).

Particle Filters

Particle Filters aim to approximately solve the filtering recursions using Monte Carlo. They are based on approximating the filtering densities by a **weighted sample**.

We will approximate $p(x_{t-1}|y_{1:t-1})$ by a set of **particles**

$$\{x_{t-1}^{(1)}, \dots, x_{t-1}^{(N)}\},$$

with (normalised) **weights**

$$\{w_{t-1}^{(1)}, \dots, w_{t-1}^{(N)}\}.$$

This means we approximate $p(x_{t-1}|y_{1:t-1})$ by a discrete distribution with value $x_{t-1}^{(i)}$ having probability $w_{t-1}^{(i)}$.

Particle Filters

How do we obtain $\{x_t^{(i)}, w_t^{(i)}\}_{i=1}^N$ from $\{x_{t-1}^{(i)}, w_{t-1}^{(i)}\}_{i=1}^N$ (and y_t)?

The idea is to plug our approximation to $p(x_{t-1}|y_{1:t-1})$ into the filtering recursions.

This gives an approximation to $p(x_t|y_{1:t})$, and we then use a Monte Carlo method, normally [Importance Sampling](#), to get weighted particles from this approximation.

Different Particle Filters differ in how they do the latter step.

Approximate Filtering Recursions

If we have weighted particles $\{x_{t-1}^{(i)}, w_{t-1}^{(i)}\}_{i=1}^N$ approximating $p(x_{t-1}|y_{1:t-1})$ we get the following approximate filtering recursions:

$$\hat{p}(x_t|y_{1:t-1}) = \sum_{i=1}^N w_{t-1}^{(i)} p(x_t|x_{t-1}^{(i)}) \quad (\text{Prediction}).$$

$$\hat{p}(x_t|y_{1:t}) \propto \hat{p}(x_t|y_{1:t-1}) p(y_t|x_t) \quad (\text{Bayes' rule}),$$

$$\hat{p}(y_t|y_{1:t-1}) = \int \hat{p}(x_t|y_{1:t-1}) p(y_t|x_t) dx_t \quad (\text{Normalising Constant})$$

The approximate predictive distribution, $\hat{p}(x_t|y_{1:t-1})$, can be viewed as:

- (i) sample $x_{t-1}^{(i)}$ from our discrete approximation to $p(x_{t-1}|y_{1:t-1})$, and
- (ii) propagate this using the Markov dynamics of the latent state.

Bootstrap Filter

The idea of Importance Sampling is to get a weighted sample from the (approximate) filtering distribution, $\hat{p}(x_t|y_{1:t})$, is:

- We choose a proposal distribution, $q(x_t)$.
- We sample particles $x_t^{(i)}$ from $q(x_t)$, for $i = 1, \dots, N$.
- We calculate the (unnormalised) importance sampling weights

$$w_t^{*(i)} = \frac{\hat{p}(x_t^{(i)}|y_{1:t})}{q(x_t^{(i)})} \propto \frac{p(y_t|x_t^{(i)}) \sum_{j=1}^N w_{t-1}^{(j)} p(x_t^{(i)}|x_{t-1}^{(j)})}{q(x_t^{(i)})}.$$

- We normalise the weights

$$w_t^{(i)} = \frac{w_t^{*(i)}}{\sum_{i=1}^N w_t^{*(i)}}.$$

Bootstrap Filter

The Bootstrap filter ([Gordon et al. 1993](#)) implements this with

$$q(x_t) = \hat{p}(x_t|y_{1:t-1}) = \sum_{i=1}^N w_{t-1}^{(i)} p(x_t|x_{t-1}^{(i)}).$$

The importance sampling weight then simplifies to $p(y_t|x_t^{(i)})$.

This choice is natural, avoids the summation over the particles at time $t - 1$ in the weight, and only requires one to be able to simulate from $p(x_t|x_{t-1})$ (rather than evaluate this density).

Bootstrap Filter

- (i) Sample $x_t^{(i)}$, $i = 1, \dots, N$ from $\sum_{j=1}^N w_{t-1}^{(j)} p(x_t | x_{t-1}^{(j)})$, by:
sampling j from $\{1, \dots, N\}$ with probabilities $\{w_{t-1}^{(k)}\}_{k=1}^N$, and then
 $x_t^{(i)}$ from $p(x_t | x_{t-1}^{(j)})$.
- (ii) For $i = 1, \dots, N$ calculate the (unnormalised) weight

$$w_t^{*(i)} = p(y_t | x_t^{(i)}).$$

- (iii) Normalise the weights, so for $i = 1, \dots, N$

$$w_t^{(i)} = \frac{w_t^{*(i)}}{\sum_{i=1}^N w_t^{*(i)}}.$$

And calculate the estimate of the conditional likelihood

$$\hat{p}(y_t | y_{1:t-1}) = \frac{1}{N} \sum_{i=1}^N w_t^{*(i)}.$$

Accuracy of Bootstrap/Particle Filter

There are two sources of error in our weighted particle approximation of $p(x_t|y_{1:t})$:

- (i) The approximation of $p(x_{t-1}|y_{1:t-1})$, and how that propagates to the error of $\hat{p}(x_t|y_{1:t})$.
- (ii) The additional Monte Carlo error from the importance sampling at time t .

The propagation of Error (i) depends on the model (how quickly it forgets the past); whereas Error (ii) depends on how we do the importance sampling.

Better particle filters try and reduce Error (ii).

Resampling/Stratified Sampling

Monte Carlo can be improved by using a stratified sample. This is primarily achieved through how we resample the particles from $t - 1$:

- Any resampling is valid provided the mean number of times $x_{t-1}^{(i)}$ is resampled is $Nw_{t-1}^{(i)}$. This leads to residual and stratified resampling (e.g. [Carpenter et al. 1999](#))
- The accuracy of one-step of the Bootstrap Filter can be measured through the [Effective Sample size](#)

$$\text{ESS} = \left(\sum_{i=1}^N (w_t^{(i)})^2 \right)^{-1}.$$

The accuracy of the importance sampling step is roughly the same as having [ESS](#) independent samples from $\hat{p}(x_t|y_{1:t})$.

Often people propagate each $x_{t-1}^{(i)}$ (and weights) only if the ESS is large.

Auxillary Particle Filters

The Bootstrap filter performs poorly if the likelihood is peaked relative to the predictive distribution. (This can particularly be a problem at time $t = 1$; initialisation issues).

- We can use a better importance sampling proposal that uses the information in y_t .
- For $t > 1$, to avoid the $O(N)$ cost of calculating the importance sampling weight we use the auxillary particle filter of [Pitt and Shephard \(1999\)](#). (This proposes on the joint space of $(x_{t-1}^{(j)}, x_t)$.)
- Using the auxillary particle filter requires we can evaluate the transition density $p(x_t|x_{t-1})$.

Auxillary Particle Filters

We have a proposal distribution of the form $\sum_{j=1}^N \beta_{t-1}^{(j)} q(x_t | x_{t-1}^{(j)}, y_t)$. But simulate j with probability $\beta_{t-1}^{(j)}$ and x_t from $q(x_t | x_{t-1}^{(j)}, y_t)$.

The importance sampling weight for (j, x_t) is

$$\frac{w_{t-1}^{(j)} p(x_t | x_{t-1}^{(j)}, y_t)}{\beta_{t-1}^{(j)} q(x_t | x_{t-1}^{(j)}, y_t)}.$$

Optimally $\beta_{t-1}^{(j)} \propto w_{t-1}^{(j)} p(y_t | x_{t-1}^{(j)})$ and

$$q(x_t | x_{t-1}^{(j)}, y_t) \propto p(x_t | x_{t-1}^{(j)}) p(y_t | x_t),$$

when you are performing IID sampling from $\hat{p}(x_t | y_{1:t})$.

Parameter Estimation

- You can do parameter estimation by adding θ to the state. This works badly in general. (Adding MCMC moves to update θ can help; [Fearnhead 2002](#)).
- You can get an unbiased estimate of the likelihood $\hat{p}(y_{1:T})$ as

$$\log \hat{p}(y_{1:T}) = \log \hat{p}(y_1) + \sum_{t=2}^T \log \hat{p}(y_t | y_{1:t-1}).$$

This can be used within pseudo-marginal MCMC ([Andrieu and Roberts 2009](#)).

- There are other ways of embedding particle filtering within MCMC ([Andrieu et al. 2010](#))
- Some approaches try to estimate the gradient of the minus log-likelihood online and perform gradient descent ([Kantas et al. 2015](#)).

Smoothing

Smoothing is a more challenging problem. There are smoothing recursions that can be solved by particle filters ([Kitagawa, 1996](#), [Godsill et al. 2004](#)).

Alternatively you can solve the smoothing problem by using a state $\tilde{X}_t = (X_1, \dots, X_t)$. But this works badly.

Fixed lag-smoothing, with $\tilde{X}_t = (X_t - L + 1, \dots, X_t)$ can work OK for small enough L .

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